Final Project Report

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Introduction

The implementation of Electronic Health Records (EHR's) has provided significant advancements in providing continuity of patient care and streamlining medical documentation. Additionally, it makes it possible for patients to quickly and easily access their records, contact their healthcare provider, and request prescriptions and follow-up appointments. One issue that such systems encounter, however, is internal security [1]. Because there are so many potential points of interaction, gatekeeping access becomes a primary concern. The use of access control is a common method to help ensure privacy in EHR systems [2]. Moreover, as the number of access points increases, so too does the attack surface. Given the extremely confidential and personally identifiable information contained within, it is of paramount importance to understand exactly the points of interaction. Therefore, the need to build an effective model and simulation of patient and medical staff interactions within a clinic can provide critical insight for maintaining core cybersecurity in the development, deployment, and implementation of a clinic EHR system. For this reason, the primary question this model seeks to simulate and investigate is how many access points are necessary for effective and efficient clinic patient flow, while also maximizing clinical cybersecurity? This is predicated on the concept that expanding access points yields a greater attack surface, thereby increasing the risk of unattended open log-in sessions, especially in healthcare where staff routinely leave and enter workstations and patient exam rooms to provide care. Additionally, each log-in point is another point of access to

the EHR system, containing confidential and federally protected information. Log-in security, user logging, and session time outs are critical aspects of maintaining access security. Still, at the core, minimizing excessive or unnecessary points of access provides an excellent foundation for all other access point cybersecurity measures and acts well as a first line of defense in protecting confidential patient information. This simulation model seeks to build a typical patient flow in a small, urgent care type clinic, and in tandem, have a parallel model run concomitantly simulating the clinical staff interactions with the clinic EHR. The purpose for this approach is to understand accurately how each step in assisting the patient causes the staff to interact with the EHR system. In so doing, it is the goal of this project to accurately simulate the standard interactions of the clinical staff with the EHR system and identify through the model the fewest possible access points the staff need to render effective patient care. In building a simulated model of this system, it will be possible to ascertain the minimum number of access points necessary while still maintaining proper and effective patient flow. The project will determine this through the modeling process and subsequent tracking of simulated user access times, that is, how long an access point remains in use and the number of user accesses at a given access point. This will allow the project to determine if a single access point is sufficient or if it causes bottlenecks in the clinic patient flow. By tracking these metrics, the project seeks to determine if a minimalist-type approach to EHR access points supports effective clinic cybersecurity, reducing the access point attack surface to a minimum, while maintaining efficient and effective clinic patient flow.

Design Methodology

In designing this model, the first aspect was to select a platform that would allow for a model study regarding the core patient flow of a typical

doctor's office or clinic. The core focus of this project is on accurately simulating patient flow from which the clinic staffs' points of access to the EHR could be ascertained. Given the focus on the core underpinnings of the functions of the clinic, aGPSS, General Purpose Simulation Software, was identified as an effective software to simulate the model, as its block diagram design is very compatible with the flow chart principle of the proposed approach and design. In building the model, the first aspect was to construct an accurate model of a typical medical clinic and translate this into a simulated model. Based on real world data, a core clinic and patient flow model was constructed in aGPSS. To focus on establishing an accurate model, the clinic was designed to work on a commonly encountered, and most basic, paradigm—a single provider model. In this model, there is one receptionist to check patients in and out, a single nurse or medical tech to triage and discharge the patients, and a single medical provider to examine the patients. Additionally, several key assumptions were made:

- All patients have access to the internet
- All patients have internet and computer literacy
- All patients have a device that supports the EHR platform
- Healthcare staff receive thorough training and use the EHR system as it was designed
- As the focus is on the EHR system and not extraneous aspects, it is assumed the clinic network infrastructure is secure
- Likewise, it is assumed the patients are using secure connections (as opposed to public Wi-Fi for example)

In the design of the model, the following metrics were identified to track statistically:

- The number of logins
- The number of logouts
- The time each user is actively logged in
- The points of access used by patients and healthcare staff

Once constructed based on this paradigm, the aGPSS clinical model was evaluated to ensure high model accuracy. Once this model had been thoroughly evaluated, a second parallel model was developed. This parallel model focused on the clinic staff's interactions with the EHR during the course of a patient's visit to the clinic, from their first interaction with the reception desk, to their final evaluation and release. In so doing, this model makes it possible to evaluate each time and point the staff need to access the EHR model. The parallel models also provide a clear relationship between the patient's journey through the clinic system, and the staff's corresponding interactions with both the patient and the EHR system, which helps ensure an accurate model. Once these were completed, the data generated was gathered and analyzed to return to the core question this model seeks to answer—how many access points are necessary for the streamlined treatment of patients and are also keep the attack surface at a minimum? Ultimately, the goal of the model is to study the potential points of access for an EHR. How many points of access are used and necessary to provide a secure, yet fully functional EHR? Ultimately, this model should provide clear proof of concept of each point of interaction between EHR and the healthcare system and patients.

Model Design

The model is built exclusively on real world data and experience. It is centered on a binary design, essentially splitting the patient flow and the staff flow into two parallel models. This allowed for ample points of cross-reference and ensured an accurate model was simulated. As such, these can be described individually.

Patient Flow

The patient flow is comprised of 11 modules, comprising 5 main states of a patient's visit: check-in, waiting, triage, examination, and discharge (check-out). Moving from GENERATE, the first module, which creates the patient inflow to the clinic, the check-in process is centered on an ADVANCE module, which simulates the variation in the check-in process. It is preceded by a SEIZE and RELEASE module respectively. Similarly, triage is centered on an ADVANCE module, also preceded by a SEIZE and succeeded by a RELEASE module. The examination stage makes similar use of the SEIZE, ADVANCE, and RELEASE modules, and is followed by a TERMINATE module to allow each patient to exit the model. The use of the SEIZE module simulates the occupation of a clinic staff member with a patient, preventing another patient from advancing to that staff member until they are free. The RELEASE module simulates the completion of the clinic member with the patient and frees them to assist the next patient.

Clinical Flow

While similar, the clinical flow model differs especially regarding staff access during the patient examination. This brings the total number of modules to 17. The 5 main stages of a patient's visit are covered by this model as well. However, it expands the EHR accesses before and after the patient examination. Moving from GENERATE, the first module, which creates the patient inflow to the clinic, the check-in process is centered on an ADVANCE module, which simulates the variation in the check-in process. It is preceded by a SEIZE and RELEASE module respectively. Similarly, triage is centered on an ADVANCE module, also preceded by a SEIZE and succeeded by a RELEASE module. The examination stage marks the major divergence from the patient model. The provider usually accesses the EHR to perform a pre-exam chart read before seeing

the patient. Here, similar use of the SEIZE, ADVANCE, and RELEASE modules is used to reflect the log-in and log-out of the provider. Following this, SEIZE, ADVANCE, and RELEASE are used again to reflect chart access by the provider for post-exam charting. This is followed by these modules again for the nurse or medical tech's login and logout to print and provide the patient with the discharge paperwork. Because all insurance, copays, and patient information are collected before treatment in the standard clinic paradigm, no further access to the EHR is needed. Therefore, the TERMINATE module is then added to allow each patient to exit the model.

Figure 1 illustrates the main design of the model in aGPSS as a block diagram.

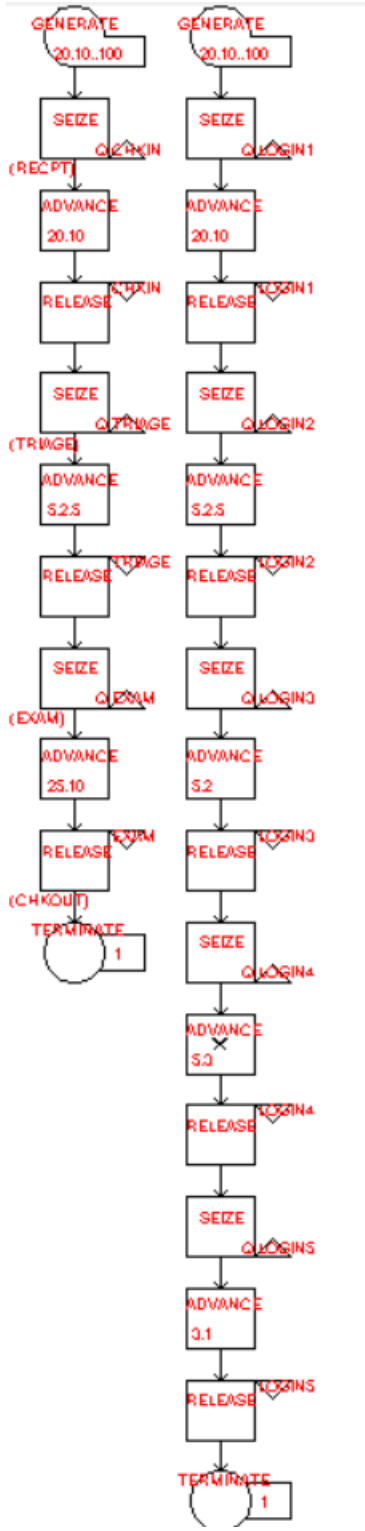


Figure 1-aGPSS Block Diagram of the Model Showing Patient(L) and Clinical(R) Models

Model Integrity & Testing

The core approach to this model has been a stochastic design. By providing a probabilistic range to each station, it allows for some degree of realistic simulation of the variability commonly encountered in medical practice. To verify the model, the reference was a flow chart based on a standard clinic model, which established the conceptual model. Although more methodical, model review and walkthrough was chosen to test the model and verify its design.

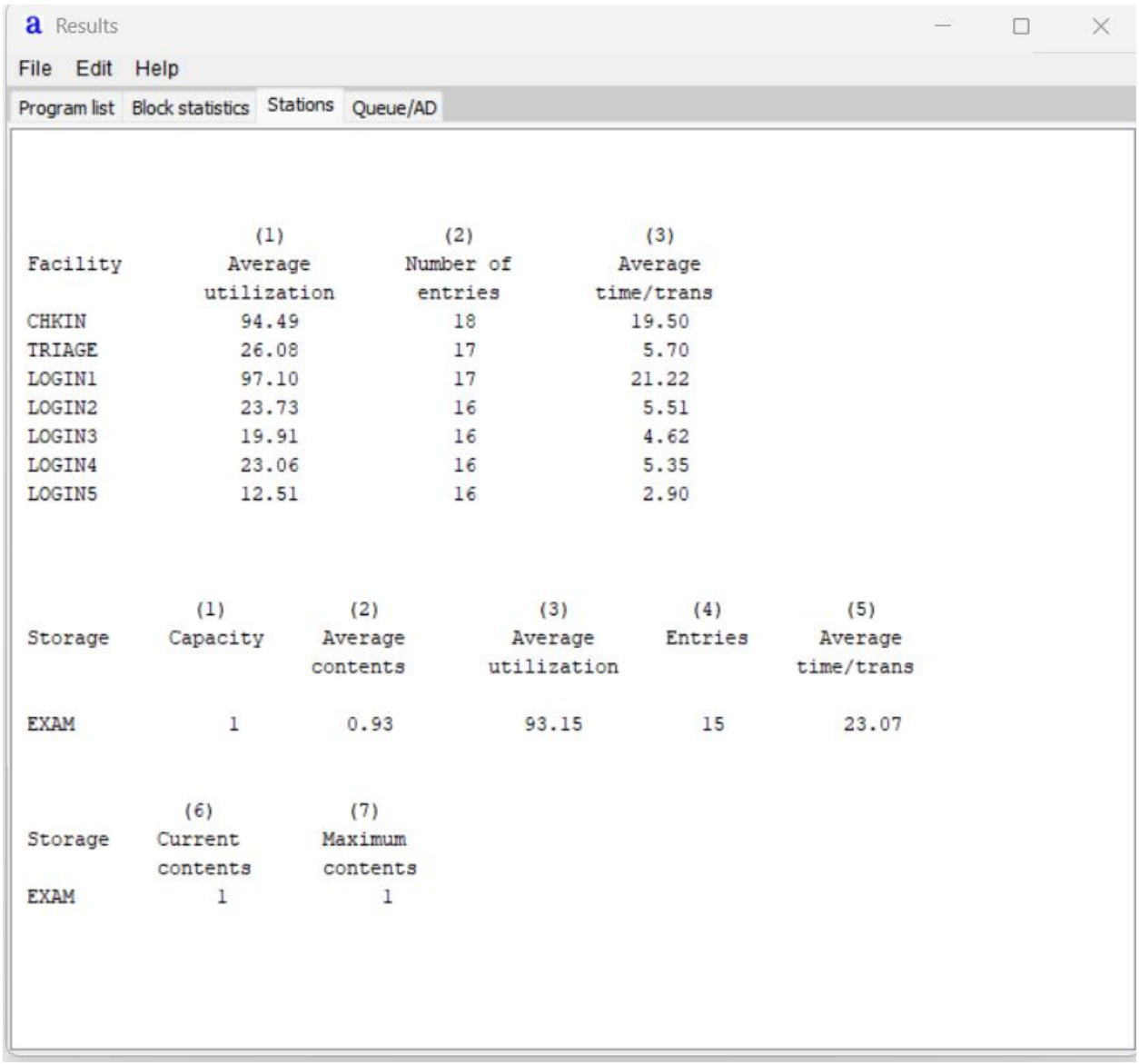
Model Verification

To verify the model, the initial parameters were considered. The primary approach for verification was a historical data comparison. Based on available data for a single provider clinic, the average number of patients seen per day can range from 20 to 50 patients, depending on the clinic type and visit type. For this model, 30 patients were selected, more reflective of an urgent care setting. Within the GENERATE module, an InterArrival time of 20 was selected, with an ½ IAT spread of 10 and a maximum number of 35 was set. For the check-in process, a delay of 20 minutes was set, with a ½ spread of 10 was used. Triage was much faster with a delay time of 5 with a ½ 2.5. For the exam, 25 was set as the delay, with a ½ of 10. These formed the basic logic for the simulation. With these values, the simulation was set to run with 30 tokens to simulate 30 patients. To verify this model, the first step was to verify that the model was stable, in that it didn't crash, and that it ran to completion. Once this was verified, the general output was evaluated to check for extreme or impossible values. Finding none, the model design and logic was then reviewed methodically module by module against a standard single provider clinic design to verify the functional logic was properly simulated in the model. This step involved tracing each module's parameters, logic, and calls to ensure the core design and logic was accurate and sound.

Model Validation

Once the model's internal design and integrity were verified, the model was then validated. To accomplish this, the model's input and output were matched against real world data to validate the accuracy of the model. The first validation was to examine the runtime. The simulation should be completed in less than 9 simulated hours, which is the usual office hours of an urgent care clinic, to be valid. If the runtime is greater than this, the number of patients or the steps simulated for them to be seen need to be re-evaluated. To perform this validation, the model was run with 30 patients and the output was validated against real world data. The results and findings are included in the next section. Following the runtime validation, the next step was to evaluate the model generated data for each station against each real-world station data. This validation step made use of the data provided by the "Stations" tab in the "Results" window post model run.

Results



Results

File Edit Help

Program list | Block statistics | Stations | Queue/AD

Facility	(1) Average utilization	(2) Number of entries	(3) Average time/trans
CHKIN	94.49	18	19.50
TRIAGE	26.08	17	5.70
LOGIN1	97.10	17	21.22
LOGIN2	23.73	16	5.51
LOGIN3	19.91	16	4.62
LOGIN4	23.06	16	5.35
LOGIN5	12.51	16	2.90

Storage	(1) Capacity	(2) Average contents	(3) Average utilization	(4) Entries	(5) Average time/trans
EXAM	1	0.93	93.15	15	23.07

Storage	(6) Current contents	(7) Maximum contents
EXAM	1	1

Figure 2-The Model Results under the "Stations" tab

The model sorts all stages, except “EXAM”, as a Facility. Average utilization shows the overall usage of a station during the simulation. CHKIN, a somewhat unfortunately named compression of “Check-In” to comply with aGPSS’s 6 character limit meshes well with LOGIN1, which is the staff aspect of the patient check-in process. LOGIN2, matches fairly closely with the utilization of TRIAGE, the patient equivalent of the two models. Both the utilization of LOGIN3, LOGIN4, and LOGIN5 are consistent with utilization expectations for these EHR accesses. This is reflected in the Average time/trans for each station, which are consistent

with the parameters set for each station. EXAM saw a utilization of 93.15, showing a high patient load. Additionally, the results show there are only 15 entries. This is 3 patients less than the number that checked in, indicating that not all the patients are seen. This is a major concern. It is a clear sign that 1 exam room is not enough.

Results

File Edit Help

Program list | Block statistics | Stations | Queue/AD

Queue	(1) Maximum contents	(2) Average contents	(3) Total entries	(4) Zero entries	(5) Percent zeros
CHKIN	3	1.13	20	4	20.00
TRIAGE	0	0.00	17	17	100.00
EXAM	2	0.50	17	4	23.53
LOGIN1	2	0.71	18	1	5.56
LOGIN2	0	0.00	16	16	100.00
LOGIN3	0	0.00	16	16	100.00
LOGIN4	0	0.00	16	16	100.00
LOGIN5	0	0.00	16	16	100.00

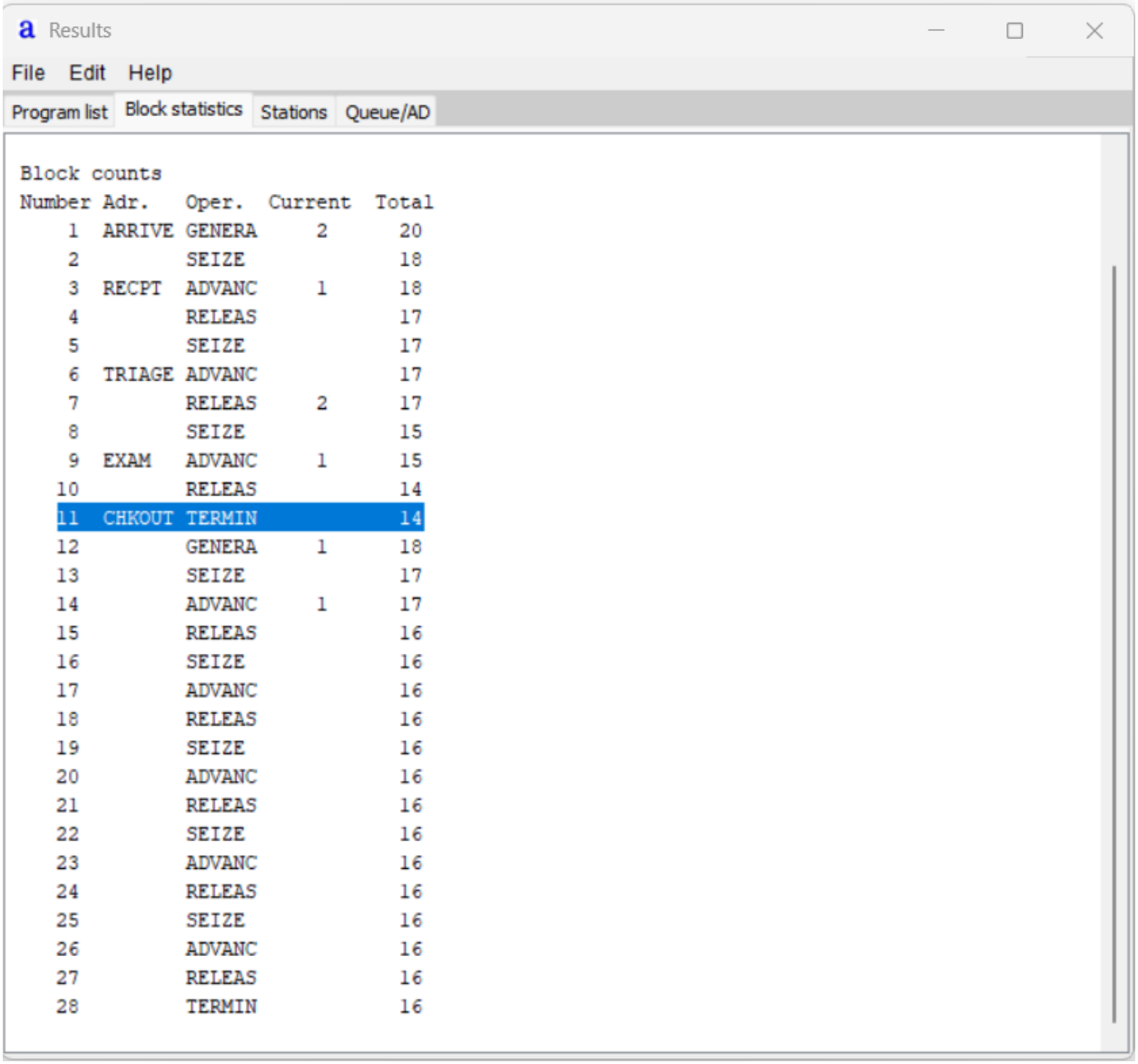
Queue	(6) Average time/trans	(7) $Average time/trans	(8) Current contents
CHKIN	20.98	26.22	2
TRIAGE	0.00	0.00	0
EXAM	10.89	14.24	2
LOGIN1	14.73	15.60	1
LOGIN2	0.00	0.00	0
LOGIN3	0.00	0.00	0
LOGIN4	0.00	0.00	0
LOGIN5	0.00	0.00	0

$Average time/trans=average time/trans excluding zero entries

Figure 3-Model Results from the Queue/AD Tab

Using the queue tracking built into the model, some interesting results are seen. The most patients waiting to check-in is 3, and yet for LOGIN1,

the clinical side of the check-in, the most patients waiting to check in is 2. Perhaps more interesting, and concerning, is that more than one patient is listed for EXAM. Based on the parameters, with a maximum capacity of 1, there should not be more than one patient in EXAM. The clearest solution here is to recheck the parameters and extend the available EXAM capacity to 4, which is a very common approach for a single provider and allows more efficient patient triage and balancing than a single room. A final note from Figure 4 is the high number of zero entries, which make bottlenecks such as check-in and exam clear from the simulation data. It is clear as well there is an unacceptable drop in the total patients seen versus total patients checked-in. Figure 5 further illustrates the initial model's need for several parameter changes.



Results

File Edit Help

Program list | Block statistics | Stations | Queue/AD

Block counts

Number	Adr.	Oper.	Current	Total
1	ARRIVE	GENERA	2	20
2		SEIZE		18
3	RECPT	ADVANC	1	18
4		RELEAS		17
5		SEIZE		17
6	TRIAGE	ADVANC		17
7		RELEAS	2	17
8		SEIZE		15
9	EXAM	ADVANC	1	15
10		RELEAS		14
11	CHKOUT	TERMIN		14
12		GENERA	1	18
13		SEIZE		17
14		ADVANC	1	17
15		RELEAS		16
16		SEIZE		16
17		ADVANC		16
18		RELEAS		16
19		SEIZE		16
20		ADVANC		16
21		RELEAS		16
22		SEIZE		16
23		ADVANC		16
24		RELEAS		16
25		SEIZE		16
26		ADVANC		16
27		RELEAS		16
28		TERMIN		16

Figure 4-The Model's Block Statistic Results

Conclusion

The initial model achieves the core goal of simulating the patient and clinic staff flow of a typical urgent care clinic and meets the primary project goal of simulating clinic staff EHR usage during patient care. The use of Block Statistics, Stations, and Queue/AD provide excellent insights into the inner workings of the model and highlight several key areas that need improvement. It is important to note, however, the model would

benefit from additional refinement particularly in the addition of three more exam rooms, and additional reasons the EHR system might be accessed, such as refilling a prescription, retrieving records for a release of records request, or to print lab results. These would add a greater dimension of accuracy to the model. Interestingly, despite these model limitations, the results indicate the largest bottleneck to a single provider clinic is having a solitary exam room—not having a single point of EHR access for each staff member. The only station this might be challenged is for the reception desk. The initial results from this model support the project hypothesis that reducing extraneous EHR access points wherever possible does not have a significant impact on patient flow, and minimizes the physical attack surface of EHR system; helping to achieve greater cybersecurity for the clinic.

References

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